# ISyE 6740 – Spring 2026
# Project Final

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Project Title: Discovering Genres by Clustering Music Based on Audio Features

## Problem Statement

Music is typically described and categorized in terms of genres such as pop, rock, and metal. These labels are often assigned subjectively, based on vague definitions, cultural acceptance, or for marketing purposes. Even within music and cultural studies, the subject of genre is hotly debated. Some argue that genre is rooted in social constructs, mostly conventions settled on by musical communities rather than any measurable musical properties (Moore, 2001); thus, genre being shaped by social rules means that music may be classified differently in different cultural settings. To add to the complexity, genre labels are often hierarchical with branching subgenres that may spawn yet even more specific subgenres. The boundaries between genres are then rarely clear or stationary.

Genres have also morphed throughout history, broadening in some cases and expanding out into subgenres in others. What is labeled pop today sounds distinct from pop music of decades past, and genres that were not used 100 years ago have since emerged. Therefore, it stands that genre labels are rarely assigned solely using measurable audio features; record labels, websites hosting music, and cultural communities all play a role in defining genres.

Understanding that genre is mostly a social construct, there may still be natural clusters in music using their audio features. I applied unsupervised machine learning methods, without genre labels, to identify groups within the dataset "Spotify Dataset 1921-2020, 600k+ Tracks," which includes audio features determined by algorithms developed by Spotify (Ay, 2021). The model results were then compared to the human assigned genre labels in the dataset to evaluate how well they reflect each other. Secondarily, I explored how audio features have shifted throughout time to include additional historical context for present trends in audio.

## Data Source

The dataset used is the "Spotify Dataset 1921-2020, 600k+ Tracks" from the user Yamac Eren Ay on Kaggle. It contains tracks extracted using the Spotify API with their associated audio features. Each track includes audio features measured by Spotify and assigned values based on their algorithms. A few audio features include *acousticness*, *danceability*, *energy*, *instrumentalness*, *liveness*, *loudness*, *speechiness*, *tempo*, and *valence*. These values are based on probabilities that Spotify assigns to each feature to describe the tracks. For example, *danceability* combines elements including tempo, rhythm stability, beat strength, and overall regularity, and *valence* describes the positiveness or negativeness conveyed by the track (Spotify, n.d.; see Appendix A). Many of these audio features are normalized by Spotify; the values are mainly scaled to be between 0 and 1. Also included in the dataset are categorical and

metadata fields such as artist name, song duration, release year, assigned genre label, and more. The dataset spans across the years 1921 to 2020 which allows for tracking how audio features have changed over time and finding musical trends. The dataset includes two files: the first file is a tracks file with features and metadata per track, and the second file is about musical artists containing genre labels for each artist.

## Methodology

Preprocessing prior to analysis began by identifying non-audio fields such as most of the metadata to not be included as inputs to the clustering algorithms. Rows that had a missing value for any of the audio features or artists were also excluded. The artists' genres were joined to the track data, and in the case where there were multiple genres assigned to artists, the first genre was kept as the label per track. All the features were then standardized using z-score normalization so that all the features with different scales, such as values for loudness and tempo, which are measured in decibels and beats per minute respectively, would not dominate distance calculations.

I applied principal component analysis (PCA) to reduce dimensionality prior to clustering. But after creating a scree plot of the variance captured by the principal components, I determined that PCA should not be used. The assessment of how well the principal components captured the variance in the data concluded in that the components did not explain sufficient variance to justify reducing the features. It took many components to capture a majority of the variance, with the first component only capturing just over 20% of the variance (see Figure 1). Therefore, PCA was not used before clustering. A 2D projection using PCA was still used to visualize clustering and provide a view of the structure of the data and cluster assignments. I instead mapped the correlation between the audio features, which showed that the feature energy was highly correlated with *acousticness* (-69%) and *loudness* (75%) (see Figure 2). *Energy* was removed to reduce the redundancy, leaving the other twelve features for clustering analysis: *acousticness, danceability, duration_ms, instrumentalness, key, liveness, loudness, modem speechiness, tempo, time_signature,* and *valence*.

Figure 1. PCA Scree Plot

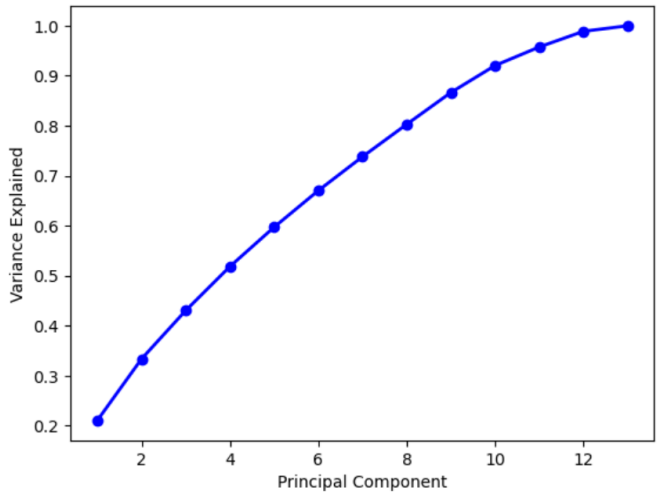
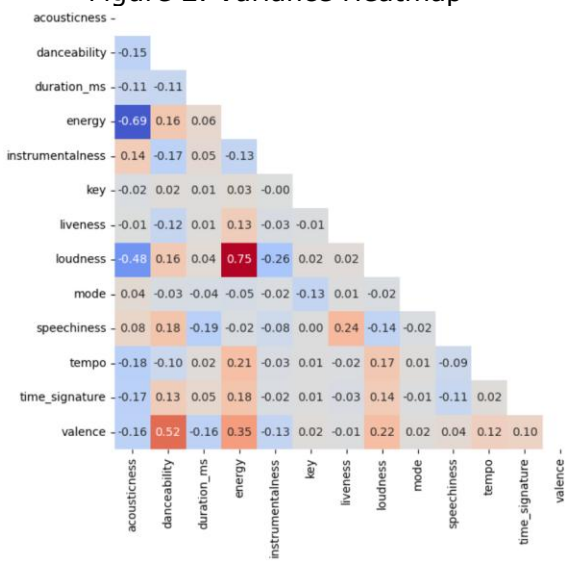
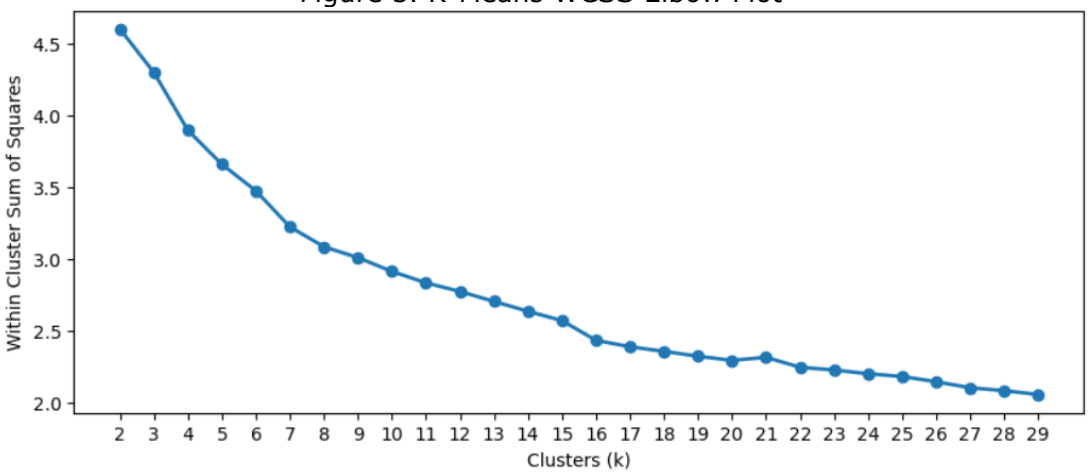



Figure 2. Variance Heatmap



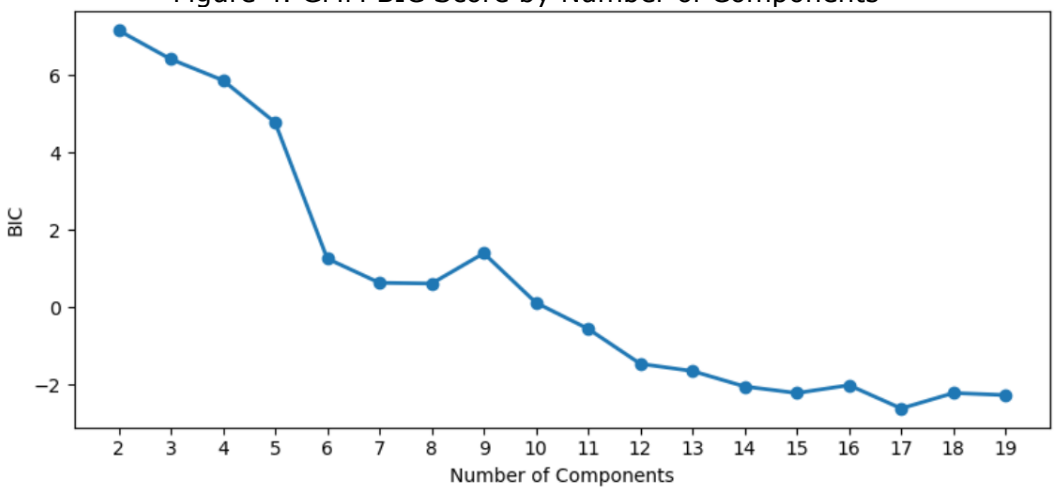
K-Means clustering was applied to the scaled audio features. The clusters were determined by minimizing the within-cluster sum of squares while iteratively assigning each datapoint to the nearest centroid and updating the centroids until no new reassignments were made. The number of clusters was selected using the elbow method where the within-cluster sum of squares was plotted along different number of clusters, *k*, and determining where the diminishing returns are not worth the additional clusters (see Figure 3). The elbow plot did not show a clear choice of *k*, as it was a gradual curve, so I ran analyses on both *k=7* and *k=15*.

Figure 3. K-Means WCSS Elbow Plot




I also performed Gaussian Mixture Models (GMM) clustering with full covariance matrices. GMM models the data as a mixture of Gaussian distributions and uses probabilistic assignments, unlike K-Means which hard assigns data points to clusters. This process can more accurately capture the inherent fuzziness of boundaries in some applications. The number of components for GMM was selected by minimizing the Bayesian Information Criterion (BIC) across a range of values. This resulted in 17 components (see Figure 4) used. The clusters generated by K-Means and GMM were then compared to the human assigned genres labels in the dataset to see how they aligned.

Figure 4. GMM BIC Score by Number of Components



## Evaluation

The K-Means and GMM cluster assignments were compared to the known genre labels in the dataset. This evaluation assessed whether tracks within a cluster tended to share a genre, or if tracks of the same genre group together in a cluster. A 2D PCA projection provided a visual representation of the cluster structure with the cluster assignments color-coded (see Figure 5 and 6). The boundaries have large overlaps, and some sections are completely mixed between the clusters and hard to distinguish. The K-Means projection is more clear than the GMM projection.

Figure 5. K-Means 2D PCA k=15

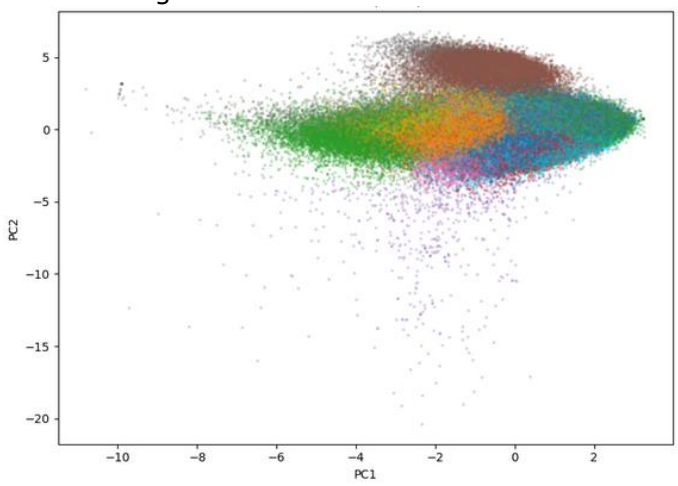
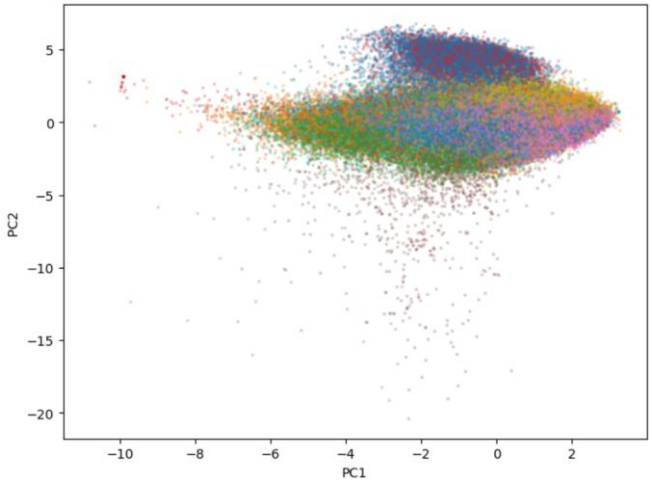



Figure 6. GMM 2D PCA Components=17




Comparisons of cluster assignment and genre included Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), Homogeneity, and Completeness. ARI measures similarity between two partitions of data by comparing cluster assignments to data labels; a value of 0 indicates a random assignment to clusters and 1 indicates a perfect match (ScienceDirect, n.d.). NMI normalizes the Mutual Information score, scaling it to be between 0 and 1, with 1 meaning a perfect correlation (scikit-learn, n.d.). Homogeneity measures whether each cluster contains only members of a single genre, and Completeness measures whether all datapoints of a genre are assigned to the same cluster.

The K-Means with 15 clusters resulted in an ARI of 0.0152 and NMI of 0.1146; suggesting low alignment between the clusters and human assigned genres, and that the given audio features do not closely replicate the genre groups. The GMM model resulted in even lower scores with an ARI of 0.0081 and NMI of 0.0977. Also suggesting low overlap between cluster assignments and genres are the Homogeneity and Completeness values. K-Means had a Homogeneity of 0.0811 and Completeness of 0.1957. GMM resulted in a Homogeneity of 0.0681 and Completeness of 0.1724. The V-measure, which penalizes imbalance between Homogeneity and Completeness, was 0.1146 and 0.0977 for K-Means and GMM respectively (see Table 1). These results imply that overall, audio features of tracks alone do not cluster into the same genres that are assigned to the musical artists.

Table 1. Comparison Results

| Method | ARI | NMI | Homogeneity | Completeness | V-measure |
|---|---|---|---|---|---|
| **K-Means** | 0.0152 | 0.1146 | 0.0811 | 0.1957 | 0.1146 |
| **GMM** | 0.0081 | 0.0977 | 0.0681 | 0.1724 | 0.0977 |

Heatmaps of the genre-clusters illustrate the spread of genres across clusters as seen in Appendix B. Genre boundaries are not clearly defined by audio features, and the majority of clusters include multiple genres. Regardless of the low alignment of clusters and genre labels, the audio feature profiles per cluster show greater distinction between groups (see Figure 7 and 8). For example, the K-Means cluster 4 is characterized by having low loudness, while the GMM cluster 5 shows low in danceability. These clusters did not align with human-assigned genres, yet they do capture tangible variation in the audio features and group tracks by their sound rather than how they are labeled.

Figure 7. K-Means Mean Features per Cluster

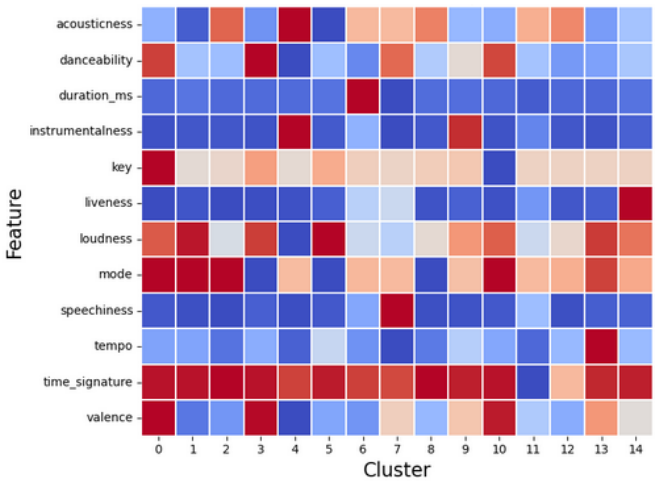
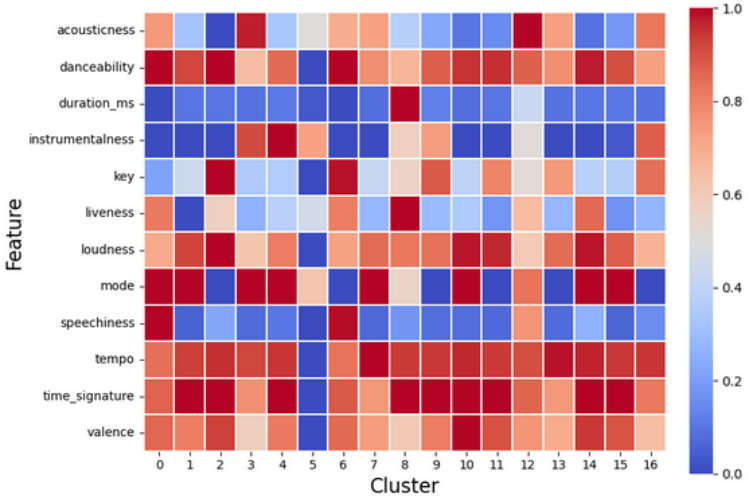
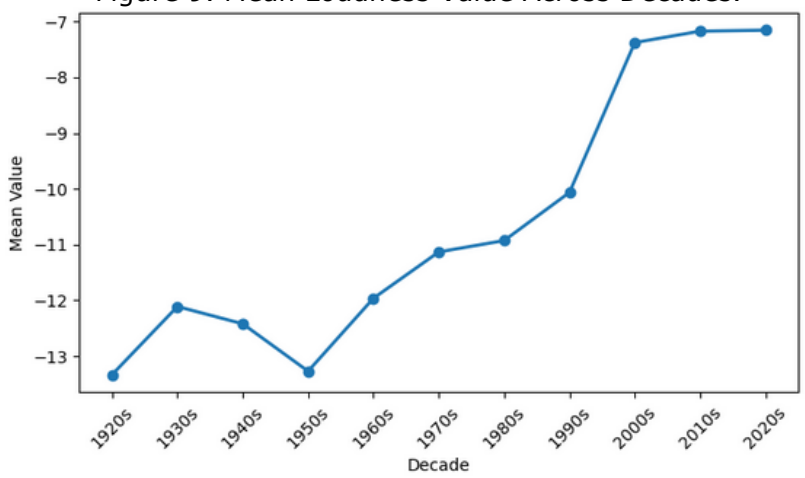



Figure 8. GMM Mean Features per Cluster




For the temporal analysis, mean values for each audio feature were plotted by decade from the 1920s through the 2010s. A few more pronounced long-term shifts include a steady decline in *acousticness* and *instrumentalness*, which follows the rise of electric production leading into the 2000s. *Energy* and *loudness* saw an increase between the 1950s and the 2000s, with the loudness surging in the 1990s in particular, dubbed the "loudness wars" (see Figure 9). More trends are seen in Appendix C. These shifts show that the audio feature landscape of tracks has changed substantially across the decades, adding context to the clustering results as tracks sharing a genre label across different time periods may sound vastly different.

Figure 9. Mean Loudness Value Across Decades.




## Discussion

Music genres assigned by humans are not well captured by audio features alone, given the clustering methods used. Both the K-Means and GMM models resulted in low alignment with the socially created genres. The ARI values indicate that the cluster assignments did only slightly better than random assignment of clusters, and the low NMI scores also reinforce the only marginal shared information between the audio feature clusters and the human decided genres. While the Completeness scores were slightly higher than the Homogeneity for both the K-Means and GMM models, it only suggests that genres may constrain audio features more than audio features determine genre. These findings support the idea that genre is largely a social construct shaped by culture, marketing, and historical conventions rather than a set of measurable audio properties. Pop tracks saw shared features such as *danceability*, *tempo*, and *valence* of genres like electronic, and the distinctions between them were not captured by clustering on the Spotify measurements.

The K-Means model slightly outperformed the GMM model across all metrics, even with GMM's probabilistic assignments and flexible covariance structure that is often better suited for overlapping and non-spherical boundaries. This was surprising in the context of musical genres. One explanation is that having twelve features may have led to overfitting even with the BIC-selected number of components. Both the BIC curve and the K-Means WCSS plots lacked a distinct elbow, so the gradual nature of the plots suggest that the data lacked strongly separated clusters, which was seen in the performance of both models. The simpler structure of the K-Means may have then been less prone to overfitting the data.

Despite the weak connection to genre labels, clustering on the audio features did provide sonic profiles that in some cases were distinct from each other, such as high *acousticness* and low *loudness* grouped together or high *danceability* with high *valence*. The unsupervised clustering shows a latent structure in music that is outside of the genre classifications. Clusters based on audio features, rather than strict genre labels, may be useful in different contexts such as recommendation algorithms.

The temporal analysis provides a historical backdrop to these findings. The audio feature trends show that the broader soundscape of music has shifted across the past century and complicates clustering tracks to genres as a track from the 1940s and a track from the 2000s may share a genre label but can sound very different from one another. One of the sharper examples is in *loudness* which had a large increase in the 1990s during the so-called "loudness wars." Driven by the increased pressure to produce more attention-grabbing and high energy music, this period saw audio producers and engineers using dynamic compressors to raise the average amplitude of recordings, essentially making the quiet parts of recordings louder and the loud parts quieter and the practice was compounded with the rise of CDs that have a defined maximum for amplitude (Abramson, 2009). The result was a concentrated change in an audio feature that did not necessarily have to do with a specific genre.

## Conclusion

The applied K-Means and GGM models clustering over 600,000 tracks spanning across a century, using twelve audio features as inputs found that the unsupervised audio-based clusters do not closely align with the human-assigned genre labels. ARI, NMI, Homogeneity, Completeness, and V-measure values were all low across both models, suggesting that audio features alone do not recover musical genre. The clusters that did emerge did provide sonic profiles of the captured audio features and showed some distinct similarities that may be useful in contexts such as recommendation systems even if those similarities did not lead to genre classification.

There were limitations to the analysis that can be expanded in future work. Genre labels were assigned at the artist level, not the track level, and there were many cases of artists having multiple genres where the analysis provided only took the first genre listed into account. Using smaller timeframes in the analysis could mitigate some of the audio feature drifts that were documented, potentially resulting in stronger cluster to genre alignments. Also, the Spotify audio features are proprietary algorithms and may not be the best predicter of the audio dimensions.

## References

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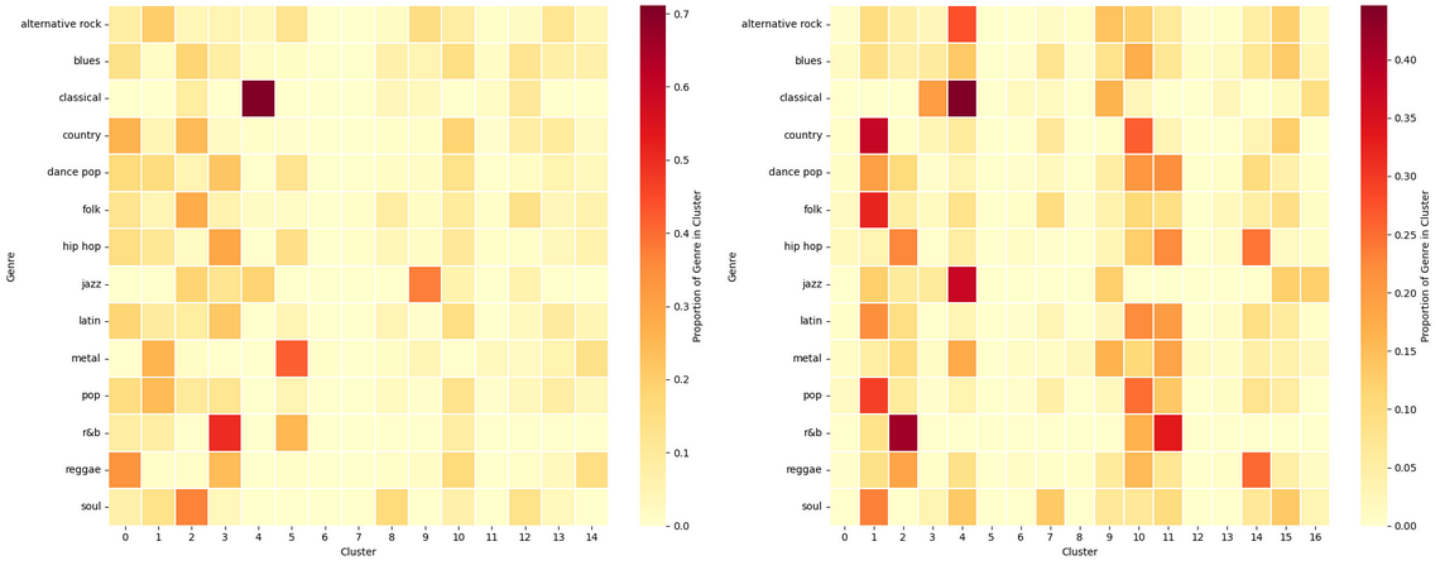
Abramson, R. (2009, December 31). *The loudness wars: Why music sounds worse*. NPR. https://www.npr.org/2009/12/31/122114058/the-loudness-wars-why-music-sounds-worse

# APPENDIX A

| Audio Feature | Description |
|---|---|
| ***acousticness*** | A confidence measure from 0.0 to 1.0 of whether the track is acoustic. 1.0 represents high confidence the track is acoustic. |
| ***danceability*** | Danceability describes how suitable a track is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity. A value of 0.0 is least danceable and 1.0 is most danceable. |
| ***durations_ms*** | The duration of the track in milliseconds. |
| ***energy*** | Energy is a measure from 0.0 to 1.0 and represents a perceptual measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy. For example, death metal has high energy, while a Bach prelude scores low on the scale. Perceptual features contributing to this attribute include dynamic range, perceived loudness, timbre, onset rate, and general entropy. |
| ***instrumentalness*** | Predicts whether a track contains no vocals. "Ooh" and "aah" sounds are treated as instrumental in this context. Rap or spoken word tracks are clearly "vocal." The closer the instrumentalness value is to 1.0, the greater likelihood the track contains no vocal content. Values above 0.5 are intended to represent instrumental tracks, but confidence is higher as the value approaches 1.0. |
| ***key*** | The key the track is in. Integers map to pitches using standard Pitch Class notation (e.g., 0 = C, 1 = C♯/D♭, 2 = D, and so on). If no key was detected, the value is -1. |
| ***liveness*** | Detects the presence of an audience in the recording. Higher liveness values represent an increased probability that the track was performed live. A value above 0.8 provides strong likelihood that the track is live. |
| ***loudness*** | The overall loudness of a track in decibels (dB). Loudness values are averaged across the entire track and are useful for comparing relative loudness of tracks. Loudness is the quality of a sound that is the primary psychological correlate of physical strength (amplitude). Values typically range between -60 and 0 db. |
| ***mode*** | Mode indicates the modality (major or minor) of a track, the type of scale from which its melodic content is derived. Major is represented by 1 and minor is 0. |
| ***speechiness*** | Speechiness detects the presence of spoken words in a track. The more exclusively speech-like the recording (e.g. talk show, audio book, poetry), the closer to 1.0 the attribute value. Values above 0.66 describe tracks that are probably made entirely of spoken words. Values between 0.33 and 0.66 describe tracks that may contain both music and speech, either in sections or layered, including such cases as rap music. Values below 0.33 most likely represent music and other non-speech-like tracks. |
| ***tempo*** | The overall estimated tempo of a track in beats per minute (BPM). In musical terminology, tempo is the speed or pace of a given piece and derives directly from the average beat duration. |
| ***time_signature*** | An estimated time signature. The time signature (meter) is a notational convention to specify how many beats are in each bar (or measure). The time signature ranges from 3 to 7 indicating time signatures of "3/4" to "7/4" time. |
| ***valence*** | A measure from 0.0 to 1.0 describing the musical positiveness conveyed by a track. Tracks with high valence sound more positive (e.g., happy, cheerful, euphoric), while tracks with low valence sound more negative (e.g., sad, depressed, angry). |

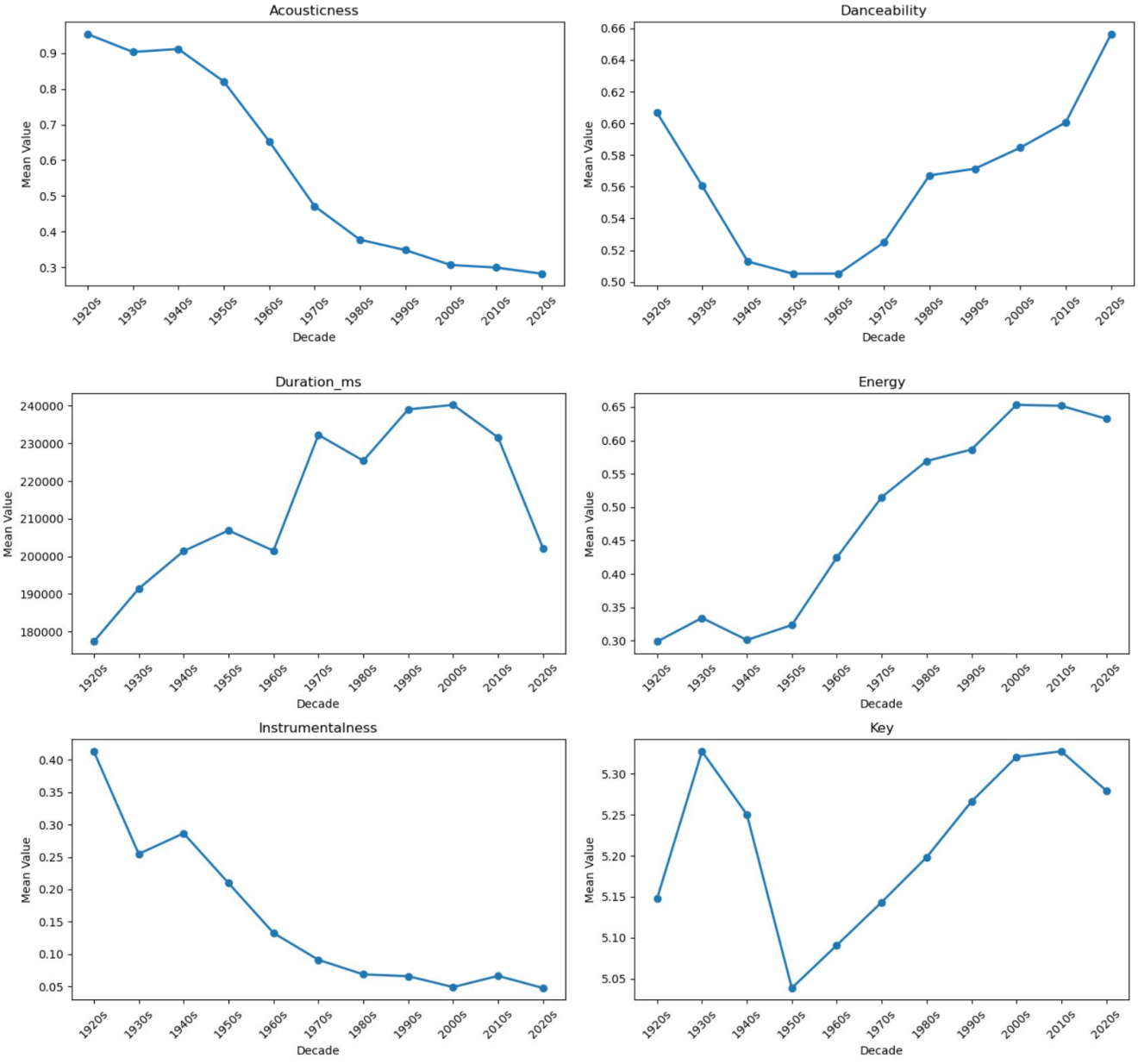
Appendix A. The set of audio features measured by Spotify.

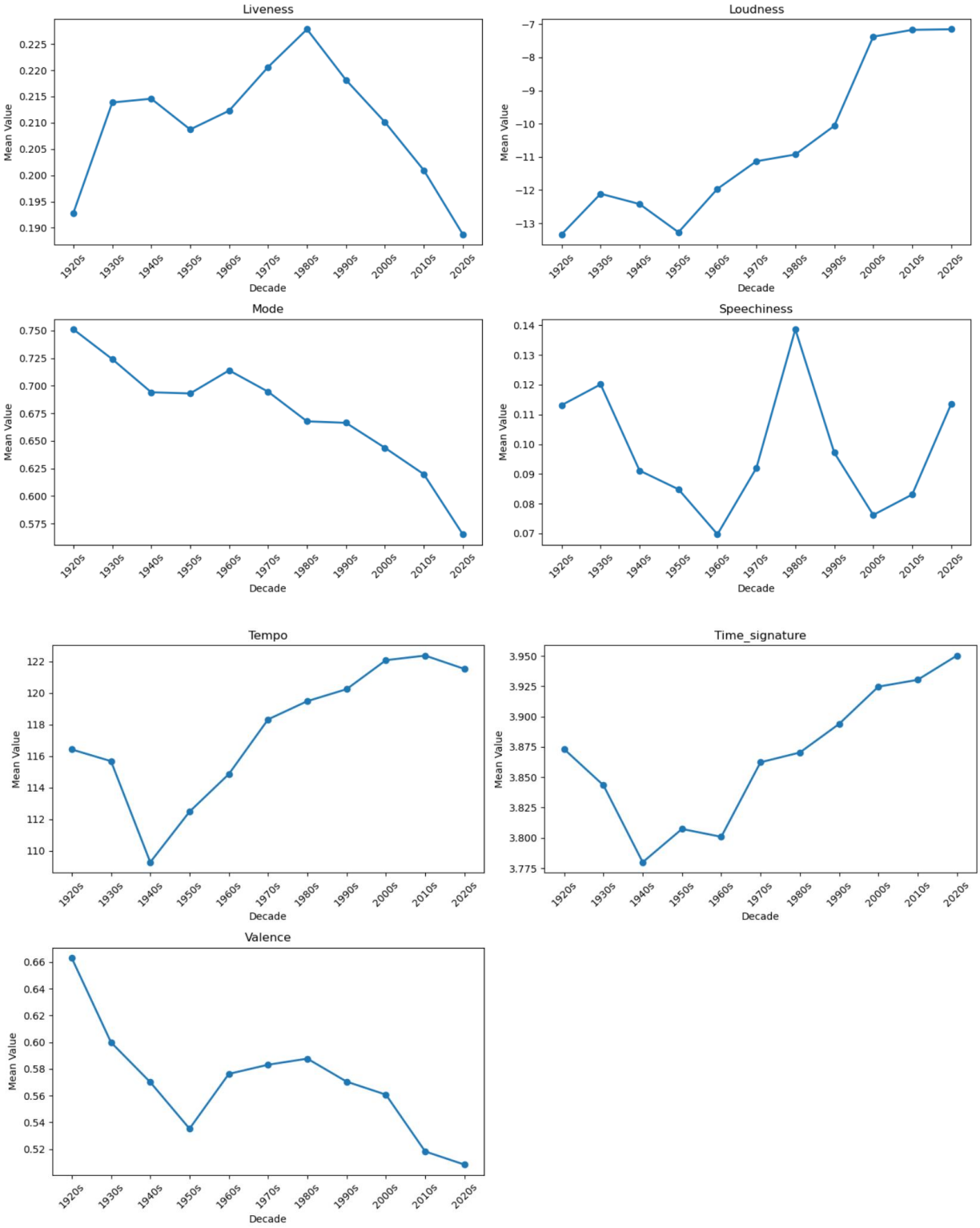
# Appendix B




Appendix B. Proportion of genre representation across clusters for K-Means k=15 (left) and GMM components=17 (right).

# Appendix C


Acousticness
Mean Value
Decade
Danceability
Mean Value
Decade
Duration_ms
Mean Value
Decade
Energy
Mean Value
Decade
Instrumentalness
Mean Value
Decade
Key
Mean Value
Decade
1920s
1930s
1940s
1950s
1960s
1970s
1980s
1990s
2000s
2010s
2020s




Appendix C. The mean values of audio features grouped by decade.